

GAURAV NIMBALKAR

Vaijapur, Maharashtra | AIMA Student | Computer Vision & Deep Learning Enthusiast

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CARRIER SUMMARY

AIMA student with strong hands-on experience in Machine Learning, Deep Learning, and Computer Vision. Skilled in developing CNN-based image classification models and data-driven AI solutions using Python, TensorFlow, and SQL. Experienced in data preprocessing, model optimization, and applied research. Seeking opportunities to build scalable, impact-driven AI systems.

ACADEMIC DETAILS

B. Tech in Artificial Intelligence & Machine Learning | 2024 - 2027 | Sanjivani University, Kopergaon

Diploma in Mechatronics | 2021 - 2024 | Sanjivai K.B.P. Polytechnic, Kopergaon

TECHNICAL SKILLS

Programming Languages: Python, C, Java

Machine Learning & Deep Learning: TensorFlow, Keras, Scikit-learn

Computer Vision: OpenCV, YOLO, MediaPipe

Data Science & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Power BI

Databases & Query Languages: SQL, Oracle

Model Deployment & APIs: FastAPI, Flask

Version Control & Collaboration: Git, GitHub

Development & Experimentation Tools: Jupyter Notebook, Google Colab

Deep Learning Concepts: CNNs, Transfer Learning, Regularization

PROJECTS

1. Fraud Detection System Using Machine Learning

- Developed a machine learning-based fraud detection system to classify financial transactions as fraudulent or genuine.
- Performed data preprocessing, feature encoding, and scaling for numerical and categorical transaction data.
- Trained and evaluated a Logistic Regression model using supervised learning techniques.
- Assessed model performance using accuracy score and confusion matrix.
- Implemented a manual input-based prediction pipeline to simulate real-time fraud detection.

Tools & Technologies: Python, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn

GitHub Link: <https://github.com/GauravN2005/Fraud-Detection-System-Using-Machine-Learning>

2. Cache-Augmented Generation (CAG) for Efficient Large Language Models (Research Project)

- Proposed and investigated a Cache-Augmented Generation (CAG) framework to enhance efficiency and reliability of Large Language Models by reusing semantically similar past responses.
- Designed a semantic caching mechanism using embedding-based similarity search to minimize redundant inference and reduce response latency in generative systems.
- Analysed cache-hit effectiveness and response quality, incorporating validation and self-repair strategies before storing generated outputs.
- Evaluated system performance through metrics such as cache hit rate, response consistency, and computational savings compared to standard generation pipelines.
- Demonstrated applicability of CAG in retrieval-augmented and conversational AI scenarios using a lightweight experimental interface.

GitHub Link: <https://github.com/GauravN2005/cache-augmented-generation-cag>

3. Aarohan - An AI-Driven Companion for Psychological Wellness

- Conceptualized and designed an AI-driven, emotion-aware virtual companion to provide instant, stigma-free emotional support through chat and video interaction.
- Developed an emotion recognition approach using facial expression analysis and text-based emotional cues to understand user mental states.
- Integrated an avatar-based interaction system to enable human-like, empathetic communication in real time.
- Designed an emotion-aware response framework focused on emotional first-aid, well-being support, and ethical boundaries.
- Evaluated the system’s potential to reduce emotional isolation and improve accessibility to mental health support without replacing professional therapy.

Domain: Artificial Intelligence, Affective Computing, Human-Computer Interaction

INTERNSHIP

Machine Learning Intern | Codec Technologies Pvt. Ltd.

Aug 2025 – Sep 2025

- Worked on CNN-based image classification as part of an AICTE & ICAC approved internship program.
- Designed and trained deep learning models using TensorFlow and Keras, achieving high validation accuracy.
- Applied data preprocessing and augmentation techniques to enhance model generalization and reduce overfitting.
- Evaluated model performance using confusion matrix, precision, recall, and F1-score.
- Gained hands-on experience with the end-to-end machine learning lifecycle.
- Trained CNN models on image datasets for multi-class classification tasks

CERTIFICATIONS

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| • Generative AI Fluency | NASSCOM (FutureSkill Prime) |
| • DevOps Fundamentals | IBM |
| • IBM Watson Studios | |
| • Predictive Modelling with IBM SPSS Modeler | |

CO-CURRICULAR ACTIVITIES

- Participated in an International Hackathon at Sandip University, Nashik.
- Attended the IEEE International Conference on Distributed Ledger Technologies (IEEE ICDLT 2025).
- Co-led the Design Team at Developer Student Club (DSC), supporting technical events and digital design activities.

PERSONAL DETAILS

Name: Gaurav Anil Nimbalkar

Date of Birth: 11/Sep/2005

Languages Known: English, Marathi, Hindi

Nationality: Indian